THE CLIENT'S PROBLEM

What I Understood

Capgemini Vertex Hub

A PLM integrator loses the first weeks of every project rebuilding the same framework, and finds inconsistencies at test, when they are expensive.

1

Every client is different

Editor (3DEXPERIENCE, Windchill, Teamcenter), vocabulary in two languages (*nomenclature de fabrication* = Plant BOM = MBOM), method (SAFe, Scrum, Waterfall).

2

Decisions live in documents

Minutes, tickets, specs, emails. A decision in one meeting silently breaks an interface agreed in another.

3

AI must be trustworthy

A generic chatbot invents facts and sends design data outside. For an industrial client, that ends the conversation.

WHAT THE CLIENT ASKS FOR

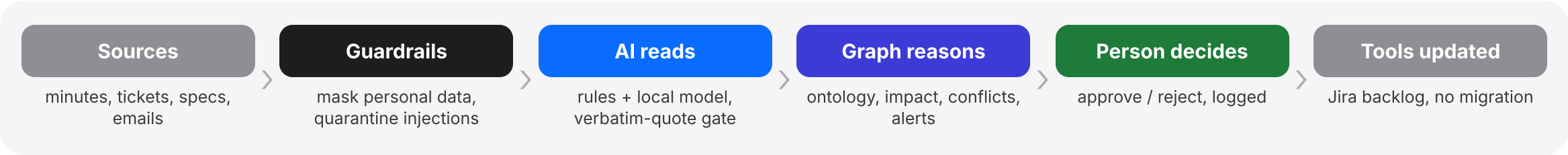
Framework in 48 h · Bilingual ontology · Detect inconsistencies · No migration · Decision support + traceability · No hallucination, no leak · Governance + billing

The case we built: ACME, an automotive supplier on 3DEXPERIENCE + SAP + MES. The PLM part description is 80 characters; the ERP field holds 40. Agreed in one meeting, contradicted in another, discovered at test. **That is the failure we prevent.**

VERTEX HUB

What I Propose

Vertex Hub: the AI reads, the graph reasons, a person decides.



PROOF ON THE DEMO PROJECT

14 days	earlier warning on the 80→40 truncation (21 Sept vs test on 5 Oct)
1 fixed, 2 new	ERP proposal: fixes INT-01, outdates TC-205, breaks a naming rule. The board sees the trade-off before approving
7·15·27	epics, capabilities, features for a new medtech client, in its method and Windchill vocabulary
35 / 1 / 0	personal data masked / injection quarantined / false alarms on project language
to measure	holdout facts found by rules + local model, every one with a verbatim source quote (rules alone: 0/4)

WHY A CLIENT ADOPTS IT

- **No migration:** reads what teams already write, writes back to Jira.
- **Nothing leaves the site:** the model runs locally.
- **Reused across clients:** catalogue, 3 editor profiles, 15 FR/EN concepts.
- **Governed:** a steward validates terms, a board approves changes, every action audited.

BILLING (assumptions)

Build	Run	Support
~€15k per client	~€1.5k per project / month	~15% of build / year

FROM BUILDING IT

What I Learned

Capgemini Vertex Hub

How a PLM project really runs

Change requests become change orders and actions; a change board decides; EBOM, MBOM and PBOM feed ERP and MES; an editor upgrade (R2025x → R2026x) is itself an impact analysis.

PLM vocabulary is the real integration problem

The same object has a different name per editor, per language and per team. Reconciling terms had to come before any AI.

The security trade-off

Our first injection filter flagged 10 of 11 normal project sentences. We chose zero false alarms over recall (12%) and said so, because a filter that blocks real minutes gets switched off.

Measure on data you never tuned on

Rules scored 8/8 on known documents and 0/4 on the holdout. Without a holdout we would have pitched a number that was false.

Local AI has limits you design around

A 7B model on a laptop needs strict JSON, a source-quote check and a cache. The guardrails make it usable, not the model size.

Keep a person in the loop

Clients trust a tool that proposes and logs. They don't trust one that changes their model on its own.

Next, if we win the contract: run on one real project's documents, connect the client's live Jira, and measure the holdout on their data before any rollout.

TEAM 103 Houssam ElMouedden · Pietro Tirante · Muhammad Ali · Govinda Bhattarai · Aymeric Slaouti · Mohamed Taleb Moustaph · Kishore Kannan Prabakaran